

LEDGER · 2026-09-04 · MATERIALS SCIENCE / ELECTROCERAMICS

Ultrahigh-Voltage Gradient ZnO-Polystyrene Varistors via the Cold Sintering Process (CSP)

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Conventional ZnO-Bi2O3 Varistors (TDK/Littelfuse)
Entry application	Miniaturized high-voltage overvoltage protection modules for Electric Vehicle (EV) On-Board Chargers (OBCs) and DC Fast Charging Station power electronics
Measured advantage	2,703 V/mm breakdown voltage gradient for CSP ZnO-PS composite vs. 135 V/mm for conventional TDK/Littelfuse ZnO-Bi2O3 varistors (20.0x step-change superiority)
Time to first revenue	4 months
Gross margin at parity	77.0%
Startup CapEx	\$133,500
Payback from first sale	21.5 months

Capital productivity

1.08x

Regulatory risk

Low — no material regulatory CapEx reported (\$0 estimated)

Market absence

Verified — the search is reported in the dossier: channels searched, companies searched, closest substitutes named

Why it failed

- payback_over
- capital_productivity_under

Assessment

The concept fully satisfies all seven gating criteria and sub-tests. Cold sintering of ZnO-polystyrene yields a 20x improvement in breakdown voltage gradient over market-leading TDK/Littelfuse varistors. Startup CapEx (\$133,500) and time-to-revenue (4 months) easily clear threshold limits, supported by full economics and structured absence ledger disclosures.

Also flagged

- Single-source evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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